Classical AI Reasoning Agent

Natural-language knowledge base → internal representation → classical reasoning → True / False

Name: SAHIL SIHAG Student ID: M26MAC008

1. Overview

The system decides whether a statement is entailed by a knowledge base written in English. Each example goes through three stages: a rule-based semantic parser builds a symbolic representation, a router picks the engine that suits it, and that engine computes the answer by classical inference. No machine-learning model or generative AI is used at any stage. The engine is plain Python with only the standard library; pandas and datasets are used for loading data.

Routing order: (1) causal language (“A causes B … Would Y occur if not X instead of X?”) goes to the counterfactual engine; (2) facts and rules go to forward chaining, parsed either as first-order rules with variables (“If someone is young then they are smart”) or as ground rules with fixed connectives (“Assuming the river is fast, we can say the house volunteers”), whichever covers more sentences; (3) loosely paraphrased rules use a lexicon-based extractor plus forward chaining; (4) any other English goes through a shallow NL→FOL parser to the resolution prover. If a test file also carries first-order formulas, those are sent straight to the prover. If nothing can be proved, the answer is False.

2. Representation

• Literals and Horn rules. A fact is a signed atom, e.g. young(bob), ¬chase(cow, cat). Adjectives become unary predicates and transitive verbs binary predicates (lemmatised: “chases” → chase). Generic subjects (someone, something, they, people, things) become a variable ?x, so “All red, cold things are round” becomes red(?x) ∧ cold(?x) → round(?x).

• Ground rules. In jise/ruletaker every clause is about a named entity and rules use five fixed connectives (If…then, Assuming…we can say, Based on…it is evident that, Given that…it follows that, When…we can conclude). Each clause becomes one propositional atom, normalised by lower-casing, dropping articles and lemmatising (“Sarah carries bags” and “sarah carries bags” both give [sarah carry bag]).

• World assumption. RuleTaker is closed-world: a statement that cannot be derived is False, and ¬p holds when p is not derivable (negation as failure). An open-world mode with classical negation is also implemented.

• Structural causal model. Each CounterBench clause becomes a Boolean equation: “A causes not B” → B := ¬A; “A and B together cause C” → C := A ∧ B; “A or not B causes C” → C := A ∨ ¬B. Variables without an equation are exogenous; “X~Bern(p)” gives one a prior, “We observed X” is evidence, and the question supplies the intervention do(X = 0).

• First-order logic. Formulas (∀ ∃ ¬ ∧ ∨ → ↔ ⊕ =) are parsed into a syntax tree, put into negation normal form, Skolemised and distributed into clauses over terms with variables and Skolem functions.

3. Reasoning algorithms

• Unification (Robinson, with occurs check) is shared by all engines.

• Forward chaining with stratified negation as failure. Predicates get strata from the rule dependency graph, so a negated predicate is complete before it is tested. Each stratum is run to a fixpoint; rule bodies are matched against an index of facts, positive literals first. Every derived fact records its rule and premises, which yields the proof trace.

• Backward chaining (SLD-style goal reduction) is implemented as a goal-directed cross-check of forward chaining; the notebook runs it on the assignment example.

• Counterfactuals by search over possible worlds (abduction → action → prediction). A depth-first enumeration of the exogenous variables keeps the worlds consistent with what actually happened; in each, the intervention overrides X and the equations are propagated forward. The answer is yes when the prior-weighted share of consistent worlds where the outcome holds exceeds 0.5.

• Resolution refutation. KB ⊨ φ iff KB ∪ {¬φ} is unsatisfiable. The prover uses set of support (only clauses descending from ¬φ are expanded), always expands the lightest clause next (uniform-cost search on clause size), and applies binary resolution, factoring, tautology deletion, duplicate and subsumption checks, a term-depth bound, and a 0.4 s / 250-clause budget per query.

4. Experiments on the public data

Dataset	Queries	Accuracy %	Majority %	Time (s)	ms / q
RuleTaker	1625	98.46	54.22	3.29	2.03
RuleTaker (classic format)	3000	99.87	50.67	10.97	3.66
CounterBench	1200	95.00	50.42	0.46	0.39
FOLIO (from English)	1204	62.21	61.79	1.77	1.47
All public queries, end-to-end	**7029**	**92.26**		**16.49**	**2.35**
FOLIO given gold FOL (engine check)	1204	81.06	61.79	24.52	20.37

End-to-end means the agent sees only the English context and statement. The last row gives the resolution prover FOLIO's hand-written first-order formulas instead of the English; it measures the reasoning engine alone and is not counted in the total. Labels are binary (True = entailed): FOLIO "False" and "Uncertain" both count as False, CounterBench "no" as False. "Majority" is the accuracy of always answering the more frequent label. Times include parsing.

Observations:

• RuleTaker. Two formats are handled: the ground-rule style (five fixed connective templates, "Facts:" / "Rules:" headers) and the classic style with variables ("If someone is young then they are smart", "Big, red things are kind"). Both are answered by forward chaining under the closed world; the table reports each source separately.

• CounterBench. Each question is turned into structural equations and answered by abduction, action and prediction. The two-step "nested" questions can be read in several ways; the reading chosen on the labelled public questions is ('set', 'set') (96.0% of the public nested questions). This choice is empirical.

• FOLIO from English is the weak point. Its sentences are long and varied ("People in this club either perform in school talent shows often or are inactive…"). The shallow NL→FOL parser rarely produces matching predicates, so accuracy stays close to the majority baseline. Translating free English into logic without a learned model is the main open problem in this design.

• FOLIO with its formulas. With the hand-written first-order formulas instead of the English, the prover is tested on its own, without the parsing step. Misses that remain can come from annotation slips (for example constants spelled differently in premise and conclusion), which no sound prover can bridge; the notebook lists a few misclassified examples.

• Prover budget. Every query gets at most 0.4 s and 250 given clauses, so that a hard FOLIO problem cannot stall the run; if no refutation is found within the budget the answer is False.

• Routing safeguards. The ground-rule parser rejects clauses containing words that need variables (someone, people, all, either …), and the paraphrase extractor only runs on short sentences. Both prevent false proofs on FOLIO stories that look superficially like RuleTaker text.

5. Reasoning traces

Assignment example (forward chaining with variables) — predicted True

Context: Bob is young. Bob is kind. If someone is young, then they are smart. If someone is smart and kind, then they are successful.

Statement: Bob is successful.

```
engine: forward chaining (CWA)
facts=4 rules=2 firings=5
query proved:
successful(bob)   [rule 1: smart(?x) ∧ kind(?x) → successful(?x)]
  smart(bob)   [rule 0: young(?x) → smart(?x)]
    young(bob)   [fact]
  kind(bob)   [fact]
```

RuleTaker (gold: True) — predicted True

Context: Facts: The globe is colorful. The globe is friendly. The globe is colorful. Rules: Given that the globe is colorful, it follows that mike is colorful. If the globe is friendly, then the dog runs marathons. Given that the city writes stories, it follows that mike is colorful. Given that the dog runs …

Statement: Mike is colorful.

```
engine: forward chaining over ground rules (CWA)
facts=6 rules=6 firings=13
query proved:
[mike be colorful]   [rule 0: [globe be colorful] → [mike be colorful]]
  [globe be colorful]   [fact]
```

RuleTaker (gold: False) — predicted False

Context: Facts: The wall dances gracefully. The river dances gracefully. The river becomes famous. Rules: Assuming the river dances gracefully, we can say the river is fast. If the wall dances gracefully, then the wall dances gracefully. Based on the river becomes famous, it is evident that the wall dances g …

Statement: The sandwich dances gracefully.

```
engine: forward chaining over ground rules (CWA)
facts=4 rules=6 firings=9
query [sandwich dance gracefully] is not derivable from the KB -> False
```

CounterBench (gold: True) — predicted True

Context: We know that Ziklo causes Blaf, Blaf and Ziklo together cause Trune, Trune causes not Vork, and Vork causes Lumbo.

Statement: Would Lumbo occur if not Ziklo instead of Ziklo?

```
engine: abduction-action-prediction over possible worlds
Blaf := (Ziklo)
Lumbo := (Vork)
Trune := (Blaf AND Ziklo)
Vork := NOT (Trune)
evidence = Ziklo
exogenous roots = ['Ziklo'] ; consistent worlds = 1
do(Ziklo=False)
counterfactual world: Blaf=0, Lumbo=1, Trune=0, Vork=1, Ziklo=0
Lumbo occurs -> yes
```

FOLIO, prover given the first-order annotations (gold: True) — predicted True

Context: All people who regularly drink coffee are dependent on caffeine. People regularly drink coffee, or they don't want to be addicted to caffeine, or both. No one who doesn't want to be addicted to caffeine is unaware that caffeine is a drug. Rina is either a student who is unaware that caffeine is a dr …

Statement: Rina doesn't want to be addicted to caffeine or is unaware that caffeine is a drug.

```
engine: resolution refutation (FOL)
⊥ (empty clause: contradiction)   [res]
  IsDependentOn(rina, caffeine)   [res]
    DrinkRegularly(rina, coffee)   [res]
      WantToBeAddictedTo(rina, caffeine)   [goal]
      DrinkRegularly(?x8270, coffee) ∨ ¬WantToBeAddictedTo(?x8270, caffeine)   [kb]
    ¬DrinkRegularly(?x8269, coffee) ∨ IsDependentOn(?x8269, caffeine)   [kb]
  ¬IsDependentOn(rina, caffeine)   [res]
    ¬Student(rina)   [res]
      AwareThatDrug(rina, caffeine)   [goal]
      ¬Student(rina) ∨ ¬AwareThatDrug(rina, caffeine)   [kb]
    Student(rina) ∨ ¬IsDependentOn(rina, caffeine)   [kb]
```

6. Limitations

• FOLIO from English: accuracy near the majority baseline. A stronger semantic parser (for instance a model that only translates English into formulas, with this prover still making the decision) would be the next step.

• The rule parsers cover the sentence patterns found in RuleTaker (attributes, relations, negation, conjunctions, If/All rules, the five connective templates); unfamiliar phrasings fall back to approximate lexicon-based extraction.

• Counterfactual answers rely on the CounterBench templates (causes / causes not / together cause / or / Bern(p) / We observed / Assume … further suppose). The reading of the two-step nested questions was chosen empirically on the public data.